

Critical Analysis: Tree of Thoughts: Deliberate Problem Solving with Large Language Models

Kai-Yu Lu

1 Methodological Strengths

1.1 Algorithmic framing of inference as deliberate search

A core methodological contribution is the explicit reframing of LLM inference as state-space search over intermediate reasoning units termed *thoughts*. This moves the decision unit from token-level continuation to semantically meaningful steps, enabling systematic branching, lookahead, and revision. The resulting methodology is specified as an algorithmic protocol rather than an informal prompting heuristic, which supports clearer attribution of performance gains to controllable inference-time computation.

1.2 Modularity enabling controlled interventions and interpretations

The approach is decomposed into separable components: thought decomposition (granularity and depth), candidate generation (proposal), candidate evaluation (value estimation or voting), and a search controller (BFS or DFS with pruning/backtracking). This modularity supports controlled changes to individual components and enables a coherent interpretation of IO prompting, CoT prompting, self-consistency, and refinement as restricted special cases under different tree-depth, tree-width, and selection rules.

1.3 Task construction and evaluation choices that isolate planning and exploration

The experimental suite emphasizes tasks that structurally require planning and exploration: arithmetic expression search (Game of 24), global coherence optimization (Creative Writing), and constraint satisfaction with backtracking (Mini Crosswords). This selection is methodologically effective because it exposes failure modes of single-trajectory decoding and tests whether explicit search and intermediate evaluation can recover from early mistakes.

1.4 Systematic comparisons, ablations, and cost reporting

A major methodological strength is the inclusion of systematic comparisons against multiple prompting baselines, paired with ablations that remove pruning and backtracking in the crossword setting. The study also reports token usage and monetary costs for selected configurations, enabling explicit performance-cost interpretation, which is particularly important for search-based inference.

Table 1: Datasets and evaluation splits (as specified in the paper). Missing fields are reported as *Not specified in the paper*.

Task	Data source	Scale / split	Evaluation target
Game of 24	4nums.com; instances ordered by human solving time from easy to hard.	1,362 total; test indices 901–1,000 (100).	Valid arithmetic equation equals 24 using each of four numbers exactly once.
Creative Writing	Inputs from randomwordgenerator.com.	100 inputs.	Four-paragraph passage ending with provided sentences; coherence is the success criterion.
5×5 Mini Crosswords	GooBix.	156 total; 20 test (indices 1, 6, ..., 96); 5 prompt (indices 136, 141, 146, 151, 156).	Correct 5×5 grid under 10 clues (5 across, 5 down).
GSM8K	<i>Not specified in the paper.</i>	100 random test questions.	Answer correctness (appendix experiment).
StrategyQA	<i>Not specified in the paper.</i>	100 random development questions.	Answer correctness (appendix experiment).

2 Key Limitations

2.1 Evaluation scope and external validity

The evaluation focuses on three deliberately constructed tasks and two additional benchmarks in an appendix. While appropriate for stress-testing deliberate search, the task set remains narrow and does not establish broad external validity for real-world deployments that involve tool use, interactive feedback, changing objectives, or long-horizon dependencies. Consequently, the study demonstrates that search can help, but does not fully specify when it is necessary, sufficient, or cost-effective across diverse task distributions.

2.2 Reliance on prompt-defined heuristics and sensitivity risk

Both proposal quality and state evaluation rely on prompt-defined behavior. This introduces sensitivity to prompt wording, decoding parameters, and model versioning. The study reports specific configurations, but does not provide a systematic sensitivity analysis over prompt variations or decoding hyperparameters beyond those used in the experiments, limiting transfer confidence under protocol perturbations.

2.3 Evaluator validity and self-reinforcement concerns

The same model class is used for generation and evaluation, enabling circular failure modes where plausible but incorrect intermediate states can be favored by self-assessment. The crossword analysis explicitly acknowledges cases where pruning can remove solvable states due to rare or obsolete words, indicating that heuristic validity can be a limiting factor. A more complete taxonomy of evaluator errors and their correlation with final failures is not provided.

2.4 Compute overhead and incomplete guidance for budget selection

Search-based inference amplifies token usage and cost (Table 6). The paper demonstrates that additional compute can yield large gains (Tables 2 and 5), but does not provide a principled policy for per-instance

Table 2: Game of 24 success rates (percentage of test instances solved).

Method	Success (%)	Methodological role
IO prompt	7.3	Direct generation baseline.
CoT prompt	4.0	Single-trajectory reasoning baseline.
CoT-SC ($k = 100$)	9.0	Multi-sample aggregation baseline.
ToT (BFS, $b = 1$)	45	Search with minimal breadth.
ToT (BFS, $b = 5$)	74	Search with bounded breadth and evaluation-guided selection.
IO + Refine ($k = 10$)	27	Iterative revision baseline using correctness feedback in this setup.
IO (best of 100)	33	Oracle-style sampling upper bound for IO under 100 trials.
CoT (best of 100)	49	Oracle-style sampling upper bound for CoT under 100 trials.

budget allocation, early stopping, or deciding when search should be invoked rather than single-trajectory prompting.

2.5 Limited robustness and failure-mode characterization

Beyond the crossword ablations (Table 5) and a qualitative discussion of pruning issues, robustness evaluations are limited. The study does not specify systematic tests for distribution shift, adversarial distractors, or prompt perturbations. Creative Writing evaluation is partly model-based and partly human-based (Tables 3 and 4), but detailed analyses of failure patterns such as plan inconsistency, local coherence versus global coherence trade-offs, or evaluator disagreement are not reported.

3 Technical Bottlenecks

3.1 Heuristic reliability as the dominant bottleneck

ToT reduces problem solving to search guided by approximate heuristics. When the heuristic is noisy or biased, BFS can retain weak branches and prune strong ones, and DFS can explore deep but unproductive paths before backtracking. The gap between ToT and the oracle “best state” output in crosswords (Table 5) indicates that the search can discover better intermediate states than are ultimately selected, revealing bottlenecks in evaluation and output-selection heuristics.

3.2 Combinatorial growth and tractability constraints

The branching factor grows rapidly with candidate count and depth. The reported instantiations keep search depth small for Game of 24 and impose immutability constraints for crosswords to keep exploration tractable. These choices reveal an underlying bottleneck: scaling deliberate search to deeper, less structured tasks requires stronger state representations, better pruning signals, and more efficient search control.

Table 3: Creative Writing evaluation (as reported).

Method	Coherency (baseline)	Coherency (+ refine)	Notes
IO	6.19	7.67	Coherency scored by GPT-4 on a 1–10 scale.
CoT	6.93	<i>Not specified in the paper.</i>	CoT refinement score is not reported.
ToT	7.56	7.91	Two-stage sample-and-vote over plans and drafts.

Table 4: Creative Writing human preference study over 100 inputs (pairwise comparison between CoT and ToT).

Comparison setting	CoT preferred	Similar	ToT preferred
Human judgments (CoT vs. ToT)	41	21	38

3.3 Irreversibility versus recoverability trade-off

Crossword solving enforces that later steps do not modify filled words or letters, improving tractability but reducing recoverability from early errors. This trade-off increases dependence on early-step proposal quality and evaluator accuracy. The sharp performance drop without backtracking (Table 5) further emphasizes the necessity of explicit recovery mechanisms for constraint satisfaction.

3.4 Proposal quality limitations not solved by evaluation alone

The reported mixed-model experiment on Game of 24 indicates that strong evaluation does not compensate for weak proposal generation. This points to a fundamental bottleneck in generating sufficiently diverse and correct candidate thoughts, not only in ranking them.

3.5 Cost-performance trade-offs without adaptive control

The approach uses fixed budgets for breadth and voting. Without adaptive control that increases or decreases search effort based on difficulty signals such as evaluator disagreement, compute can be wasted on easy instances and remain insufficient for hard ones. This bottleneck constrains practical deployment under latency and budget limits.

4 Research Implications

4.1 Inference-time computation and control as capability levers

The empirical results show that non-trivial gains can be achieved by changing inference-time computation and control without changing model parameters. This supports the view that capability is shaped by both the base model and the query protocol, and motivates benchmark reporting standards that include inference-time budgets and controller details.

Table 5: 5×5 Mini Crosswords results and ablations (percentages).

Method	Letter	Word	Game	Interpretation of variant
IO	38.7	14.0	0	Direct generation under constraints.
CoT	40.6	15.6	1	Linear reasoning with intermediate text.
ToT (DFS + prune + backtrack)	78.0	60.0	20	Search with constraint checking and recovery.
ToT + best state (oracle output)	82.4	67.5	35	Upper bound given explored search states.
ToT - prune	65.4	41.5	5	Removes evaluator-guided pruning.
ToT - backtrack	54.6	20.0	5	Removes recovery from early commitments.

Table 6: Reported token usage and cost per case (as specified in the paper).

Task	Method	Completion tokens	Prompt tokens	Cost (USD)
Game of 24	IO (best of 100)	1.8k	1.0k	0.13
Game of 24	CoT (best of 100)	6.7k	2.2k	0.47
Game of 24	ToT (BFS, $b = 5$)	5.5k	1.4k	0.74
Creative Writing	IO	0.9k	0.4k	0.06
Creative Writing	CoT	0.9k	0.4k	0.07
Creative Writing	ToT	4.0k	2.9k	0.32

4.2 LLMs as heuristic providers for algorithmic controllers

Using an LLM as both generator and heuristic evaluator provides a bridge between classical search and foundation models. This implies that progress can arise from improved interfaces between generative priors and algorithmic controllers, including better state abstractions and verification-aware evaluation signals.

4.3 Protocol design as part of system specification

Performance depends on decomposition, proposal prompts, evaluation prompts, search budgets, and selection rules. This implies that comparisons across methods should treat the protocol as part of the system specification and report it with the same rigor typically reserved for training configurations.

4.4 Reliability constraints of self-evaluation

The pruning discussion and the best-state oracle gap in crosswords highlight reliability limits of self-evaluation. These findings imply that high-stakes deployments require external verification, calibration, and conservative pruning policies to prevent systematic elimination of correct solutions.

5 Potential Research Directions

5.1 Verifier-augmented evaluation for structured tasks

Incorporate external verifiers to complement self-evaluation where formal correctness is available. Arithmetic tasks can use symbolic checking. Crossword constraints can be validated with lexicon constraints

and clue-answer consistency checks where applicable. This direction directly targets the evaluator validity limitations evidenced by incorrect pruning.

5.2 Calibration and uncertainty-aware pruning

Develop evaluation protocols that quantify uncertainty from vote disagreement and use uncertainty-aware pruning rules. Conservative pruning under high disagreement can reduce false eliminations and improve robustness. Repeated evaluation with diversified prompts can be used to stabilize ranking signals.

5.3 Adaptive search control under explicit budgets

Introduce budget-aware controllers that allocate breadth, depth, and evaluation calls per instance based on difficulty signals. Early stopping and dynamic beam sizing conditioned on evaluator entropy can improve cost-effectiveness relative to fixed budgets.

5.4 Learning improved proposal distributions and decompositions

Fine-tune models to generate higher-quality and more diverse candidate thoughts under ToT-style supervision, and learn decomposition policies that decide when to operate at coarse planning granularity versus fine-grained step expansion. This direction targets the proposal-generation bottleneck highlighted by mixed-model evidence.

5.5 Scaling to deeper tasks via richer state representations

Develop stronger state representations and constraint propagation mechanisms for long-horizon tasks, including explicit tracking of commitments, structured memory, and incremental verification. These components can reduce effective branching and mitigate drift, improving scalability of deliberate search.

5.6 Robustness-focused evaluation suites for deliberative inference

Construct standardized stress tests for deliberate inference, including prompt perturbations, misleading intermediate candidates, and distribution shifts. Metrics should separate failures of proposal generation, evaluation, and search control to enable targeted improvements.

6 Conclusion

The paper provides a clear methodological contribution by formalizing deliberate LLM inference as search over intermediate thoughts guided by self-evaluation. Strengths include algorithmic clarity, modularity enabling controlled comparisons, mechanism-aligned ablations, and explicit reporting of performance-cost trade-offs. The most critical limitations are the narrow evaluation scope, substantial compute overhead, and reliance on prompt-defined heuristics whose failure modes are only partially characterized. The dominant technical bottlenecks are heuristic reliability, combinatorial growth, irreversibility constraints, and proposal generation quality. The most actionable research directions are verifier-augmented evaluation, calibrated and uncertainty-aware heuristics, adaptive budgeted search control, and learning improved thought proposal and decomposition strategies.